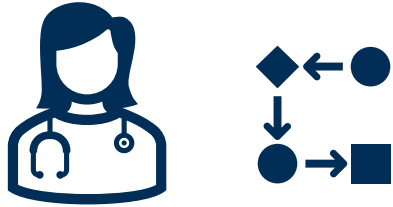


# Personal Genomic Data Sharing:

## Current Challenges & Future Possibilities

# The Main Challenges (for clinical use)

Clinical Workflow Integration



The Regulatory Landscape



The Genomic Data



Technical Implementation



# Genomic Data for Clinical Use

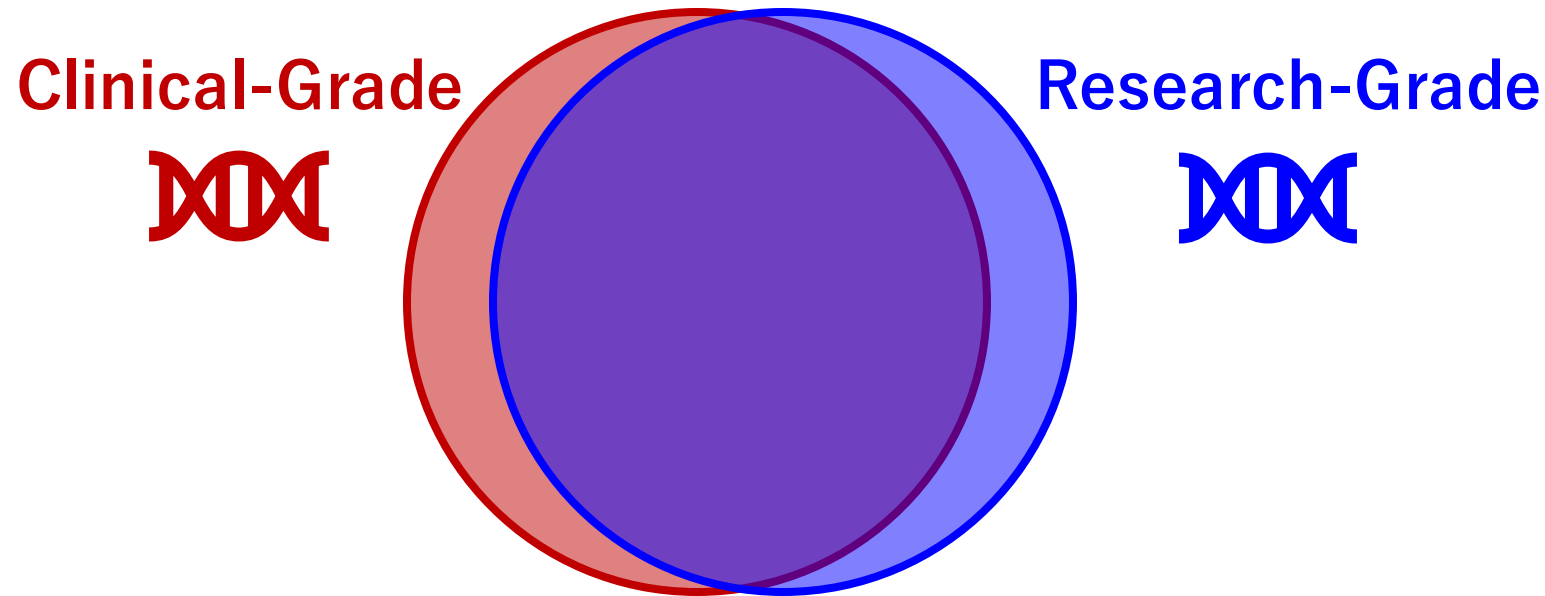
-  Format is use-case specific
-  Whole Genome Sequence (WGS) data
-  Large Projects: 1+ Million Genomes (EU) <sup>1</sup>



<sup>1</sup> <https://digital-strategy.ec.europa.eu/en/policies/1-million-genomes>

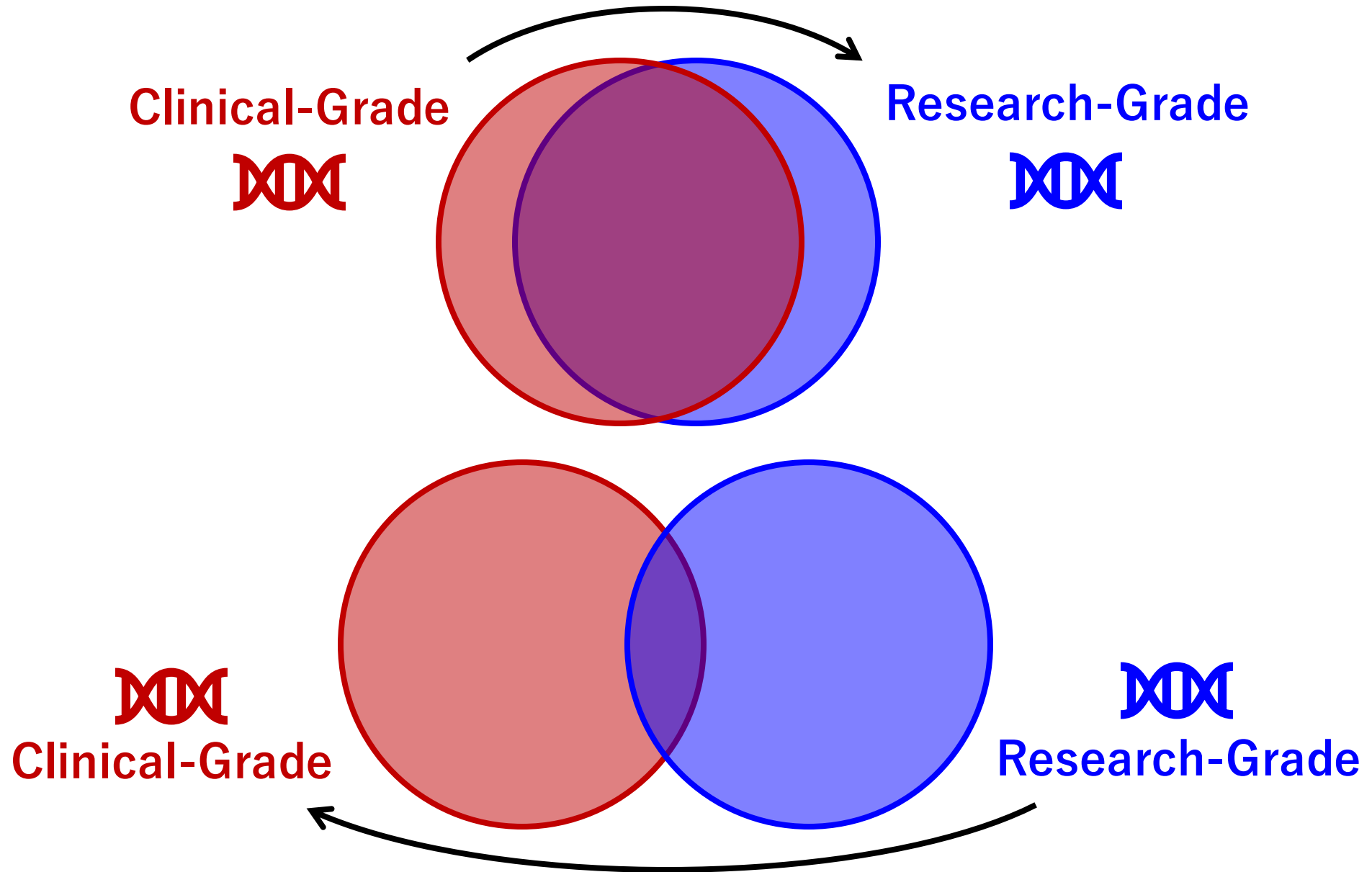
# Genomic Data for Clinical Use (assumption)

-  Format is use-case specific
-  Whole Genome Sequence (WGS) data
-  Large Projects: 1+ Million Genomes (EU) <sup>1</sup>



<sup>1</sup> <https://digital-strategy.ec.europa.eu/en/policies/1-million-genomes>

# Genomic Data for Clinical Use (reality)



# The current landscape

Intended use is VERY important

Clinical use  $\neq$  Research use

**\*\* Funding goes predominantly to RESEARCH-GRADE \*\***

Cheaper

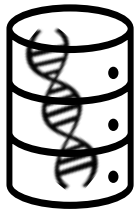
Fewer regulatory challenges

Funders / Policy makers misunderstanding

**\*\* But how do we “translate” to clinical use?? \*\***

# Sharing Genomic Data – Technology Considerations

## Key Implementation Factors



Data Storage

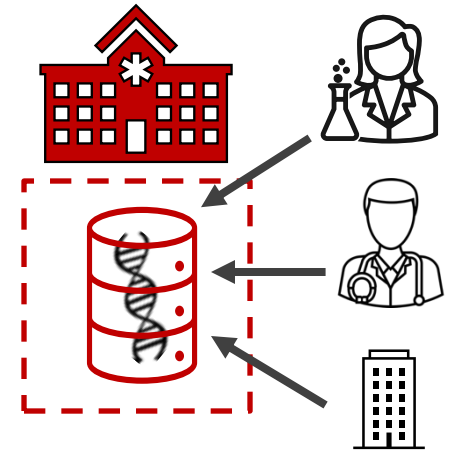
Considerations:

- patient consent
- data security
- access controls

&

Considerations:

- consent changes
- governance
- infrastructure

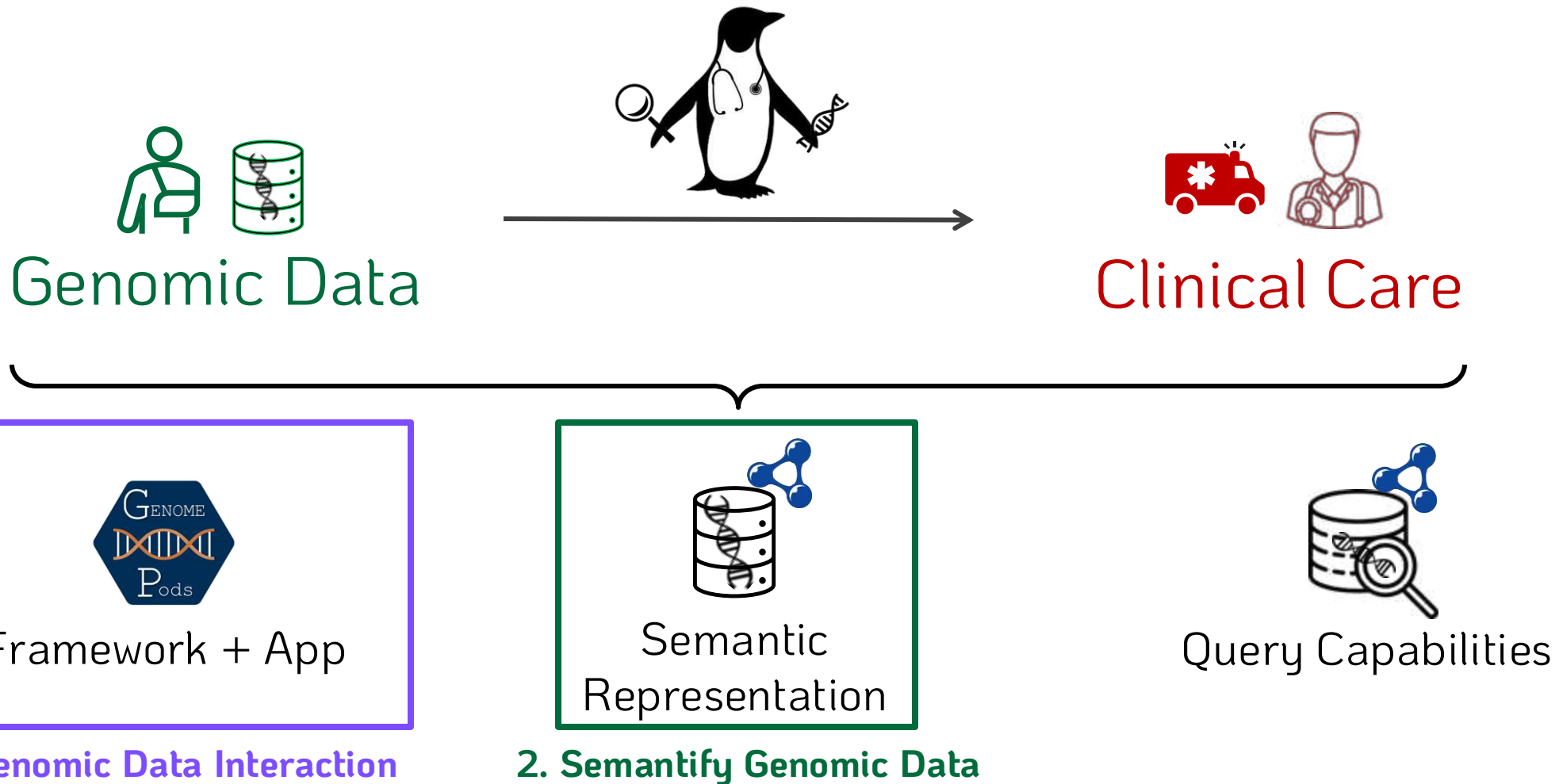


Data Sharing

A more in-depth analysis of these challenges + potential solutions detailed  
in an upcoming publication in *Computers in Biology and Medicine*

# PENGQUIN:

PERsoNal Genome QUery IN  
healthcare and clinical practice





# 1. Data Interaction Framework

*A Web Application to:*

 *Upload Data to a Solid Pod*

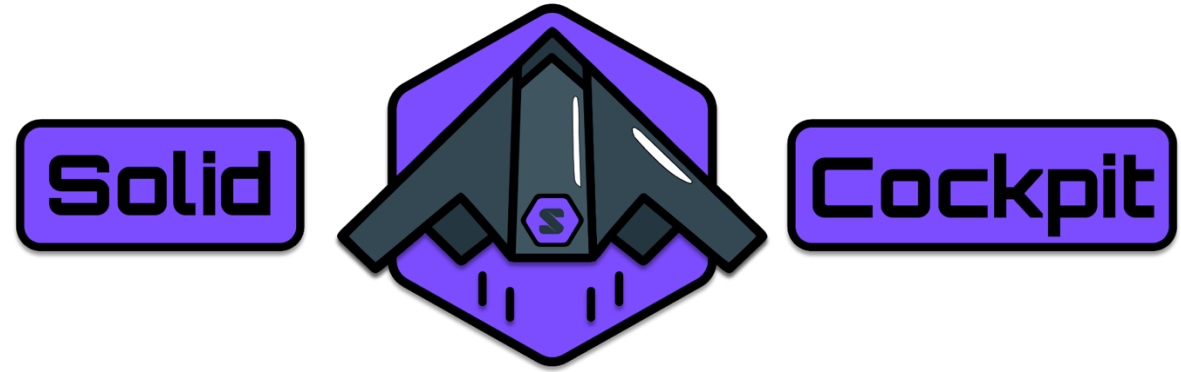
 *Browse contents of a Solid Pod*

 *Edit Privacy of (and share) data in a Solid Pod*

 *Query Data in Solid Pods + SPARQL Endpoints*

 *Includes a query cache for storing queries and results*

*\*\* Enables patient, provider, and auditor control over data \*\**

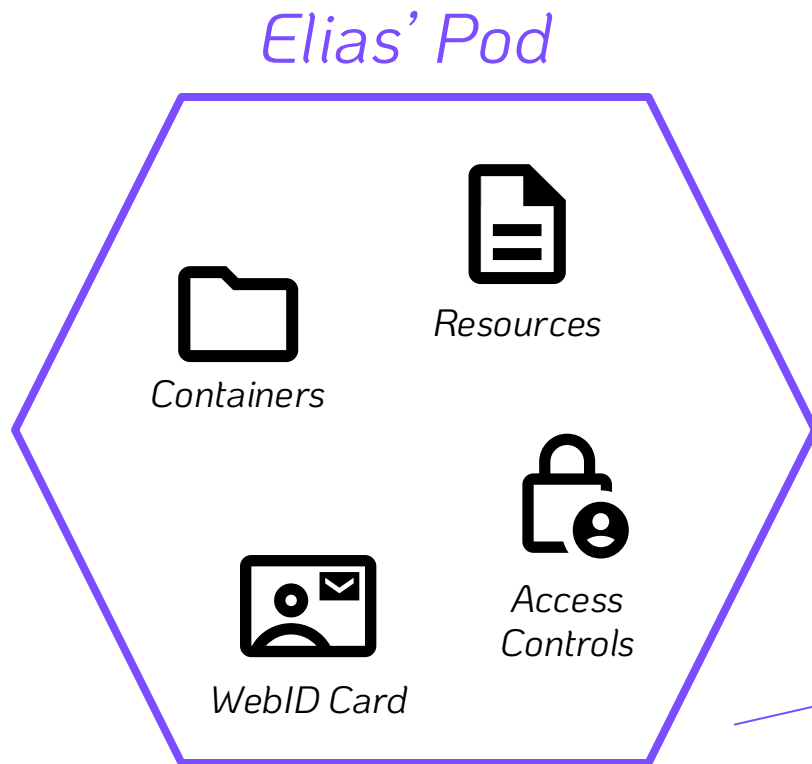


# What is a Solid Pod?



*Secure, Decentralized Data Storage Specification*

- ↳ *Pod exists on a Server (accessible to the Web)*
- ↳ *Can store files, RDF data, and Linked Data*
- ↳ *Provides control policies for stored data*



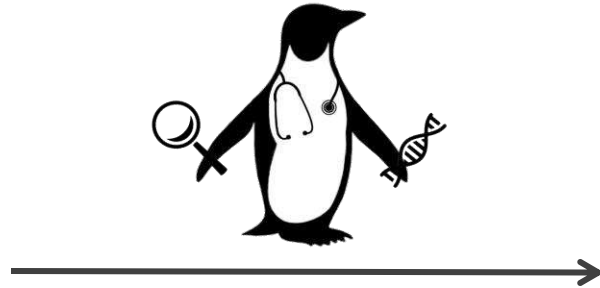
**\*\* Enables secure, authenticated sharing via HTTP protocols \*\***

# PENGQUIN:

PERsoNal Genome QUery IN  
healthcare and clinical practice



Genomic Data

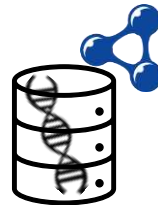


Clinical Care



Framework + App

**1. Genomic Data Interaction**



Semantic  
Representation

**2. Semantify Genomic Data**



Query Capabilities

# 2. Genomic Data Representation

From a file-based approach

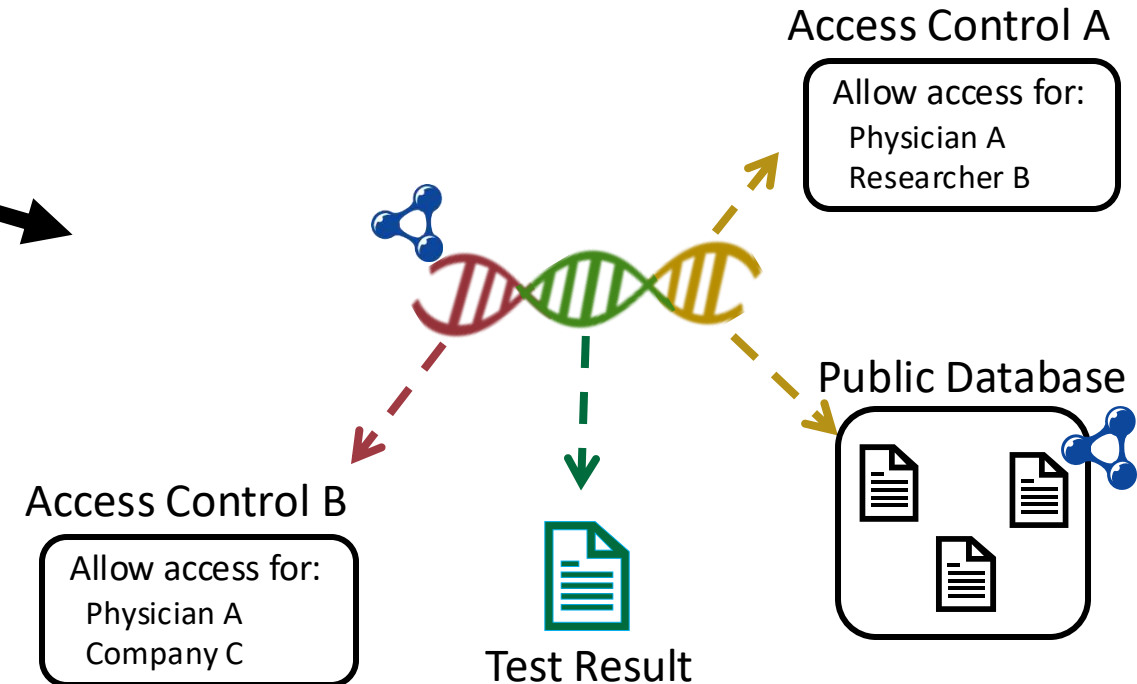


To a granular, dynamic RDF approach

Conversion using existing ontologies

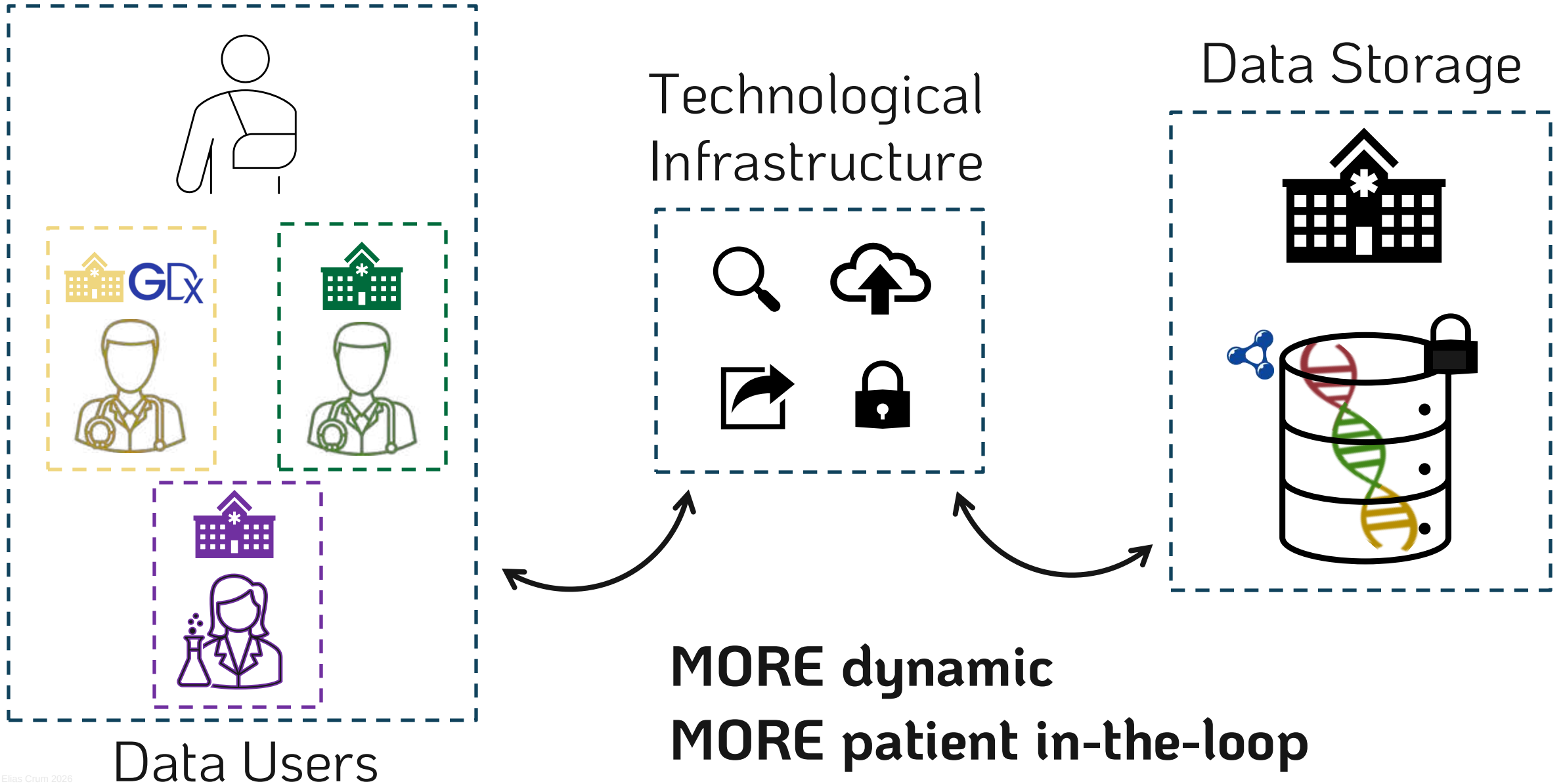
Policy definition via ODRL<sup>1</sup>

Implementation within Solid Pod Env



<sup>1</sup> <https://www.w3.org/TR/odrl-model/>

# Future Possibilities?



# Thank you!



More PhD specifics



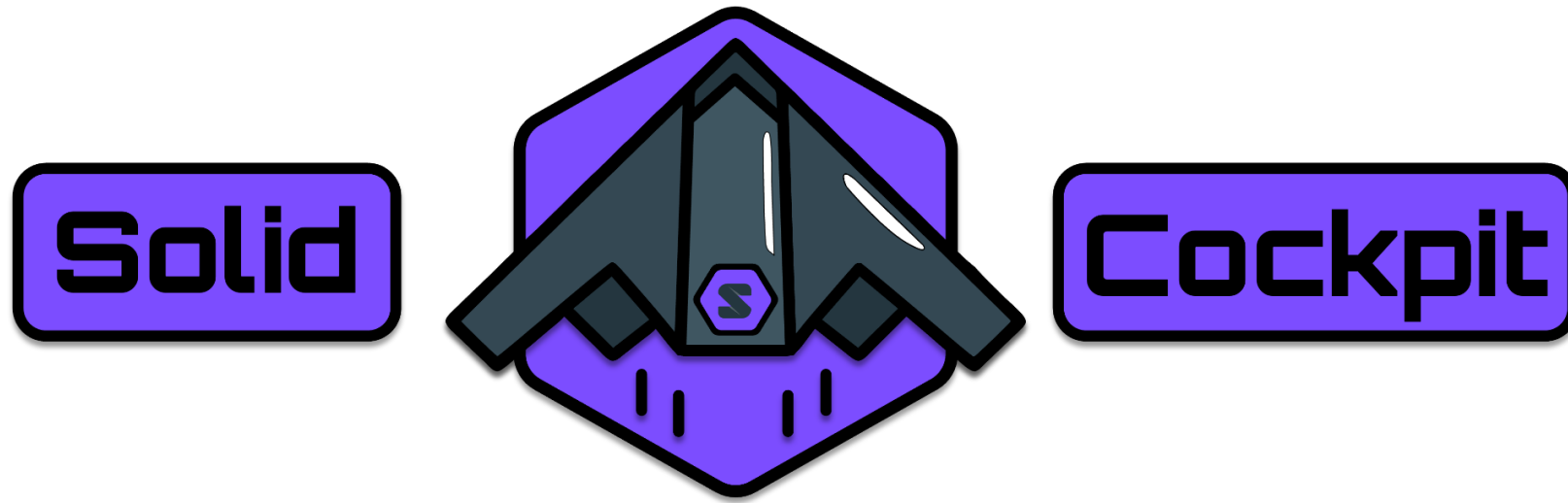
LinkedIn



GitHub

# Informal Demo / Tutorial Later Today

(for those interested)



[knowledgeonwebscale.github.io/solid-cockpit/](https://knowledgeonwebscale.github.io/solid-cockpit/)

# Supplemental Slides



# Genomic Data

## Clinical-Grade

Higher read quality

More expensive

Stricter use-regulations

## Research-Grade

Lower read quality

Less expensive

Flexible use-regulations

| Dimension                        | Clinical Grade Genome Data Sharing                           | Research Grade Genome Data Sharing                       |
|----------------------------------|--------------------------------------------------------------|----------------------------------------------------------|
| 1. Primary Purpose and Scope     | Patient care and medical decision-making.                    | Discovery and hypothesis-driven studies.                 |
| 2. Data Accuracy and Quality     | Stringent quality control (QC) and validation.               | High but no absolute QC standards.                       |
| 3. Regulatory Requirements       | Subject to medical regulatory frameworks.                    | Subject to research ethics regulations.                  |
| 4. Consent Requirements          | Clinical-informed consent.                                   | Research-informed consent.                               |
| 5. Privacy and Confidentiality   | Highest level of patient privacy required.                   | Strong but variable privacy measures.                    |
| 6. Security Controls             | Robust, enterprise-grade security.                           | Secure but more flexible.                                |
| 7. Data Governance and Oversight | Hospital/clinic-based governance.                            | Institution or consortium-based governance.              |
| 8. Ethical Considerations        | Focus on direct patient benefit and “do no harm.”            | Focus on scientific advancement and participant welfare. |
| 9. Infrastructure Requirements   | Highly reliable, clinically validated systems.               | Flexible, scalable research infrastructure.              |
| 10. Architecture Requirements    | Standardized, interoperable clinical systems.                | Flexible, modular architectures.                         |
| 11. Legal Liabilities            | High.                                                        | Moderate.                                                |
| 12. Interoperability             | High necessity for clinical integration.                     | Variable.                                                |
| 13. Data Ownership and Control   | Patient-oriented ownership with institutional custodianship. | Institution or sponsor-owned.                            |
| 14. Data Sharing Scope           | Limited to necessary clinical stakeholders.                  | Broad or consortium-based.                               |
| 15. Cost and Funding Models      | Healthcare reimbursement or institutional budgets.           | Grant-funded or institutional research budgets.          |

# Genomic Sequence Data Sharing for Clinical Practice: A Scoping Review



**Aim:** Identify existing implementations, guidelines, and challenges of clinical sharing of genomic data.

**Methods:** Reviewed 55 studies, 57 national genomics projects, and 20 major genomics companies.

**Results:** Clinical genome sharing implementations exist, yet scalability and widespread adoption remain limited.

**Suggested steps forward:**

Harmonize legal bases for  
secondary clinical use of  
research data



Mandate (GA4GH)  
technical standards



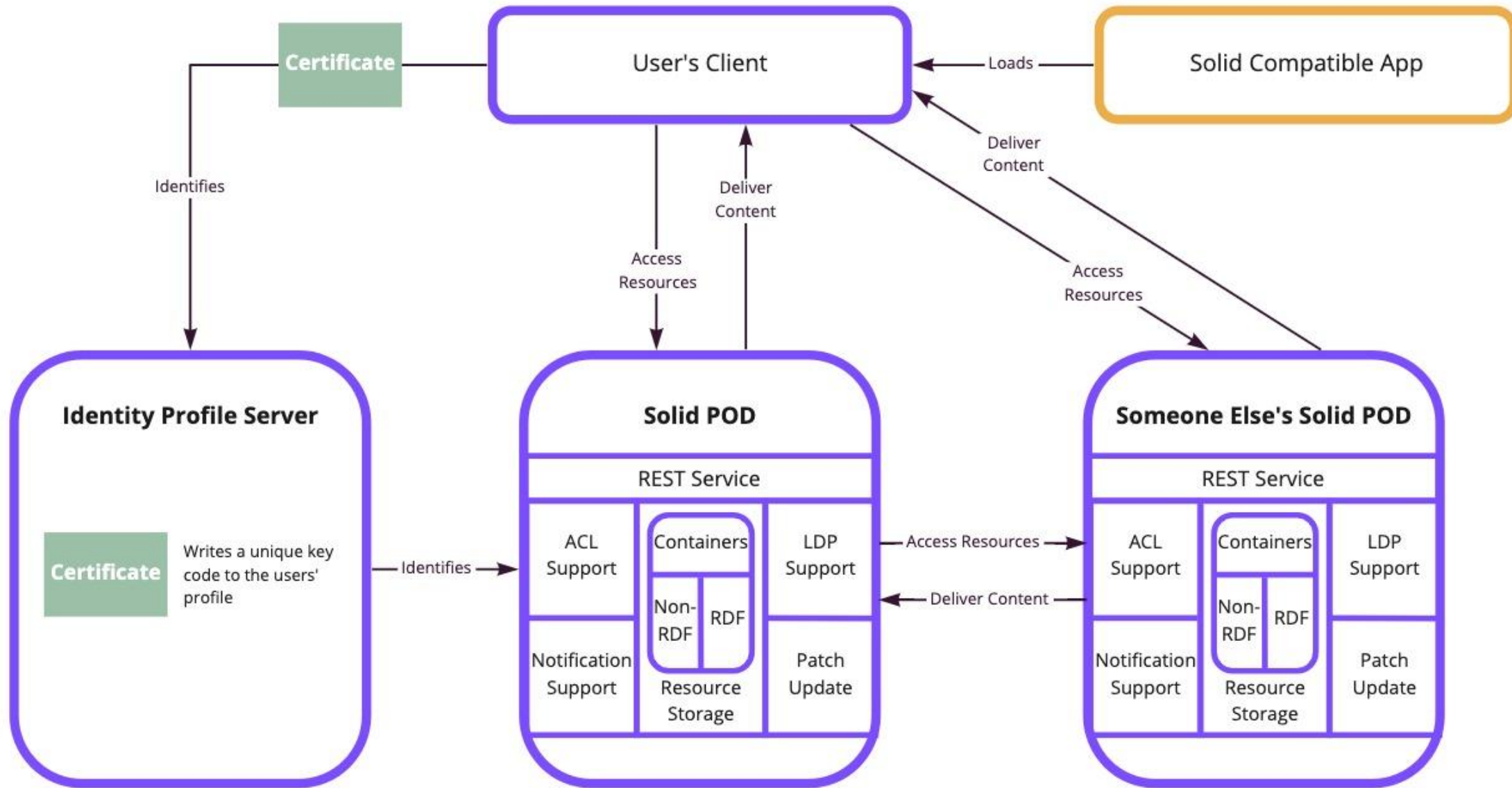
Scale laboratory  
accreditation & continual  
pipeline validation



Align economic incentives



# Solid's Architecture



miro

# Solid Pod Implementations?



BBC MyPDS App



Flemish Data Utility Company

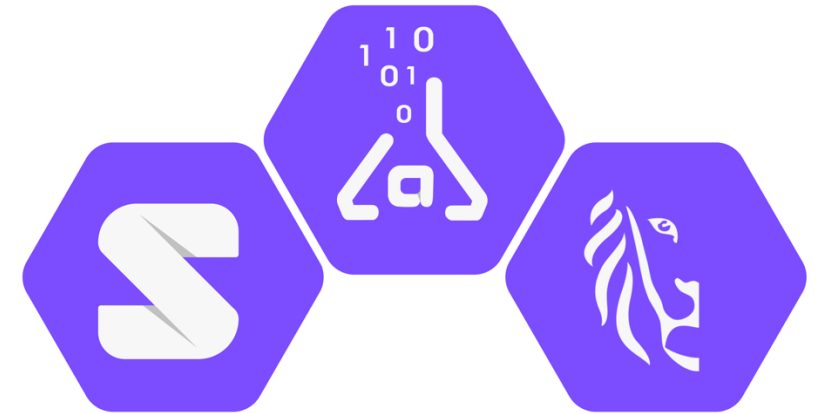


health data vaults



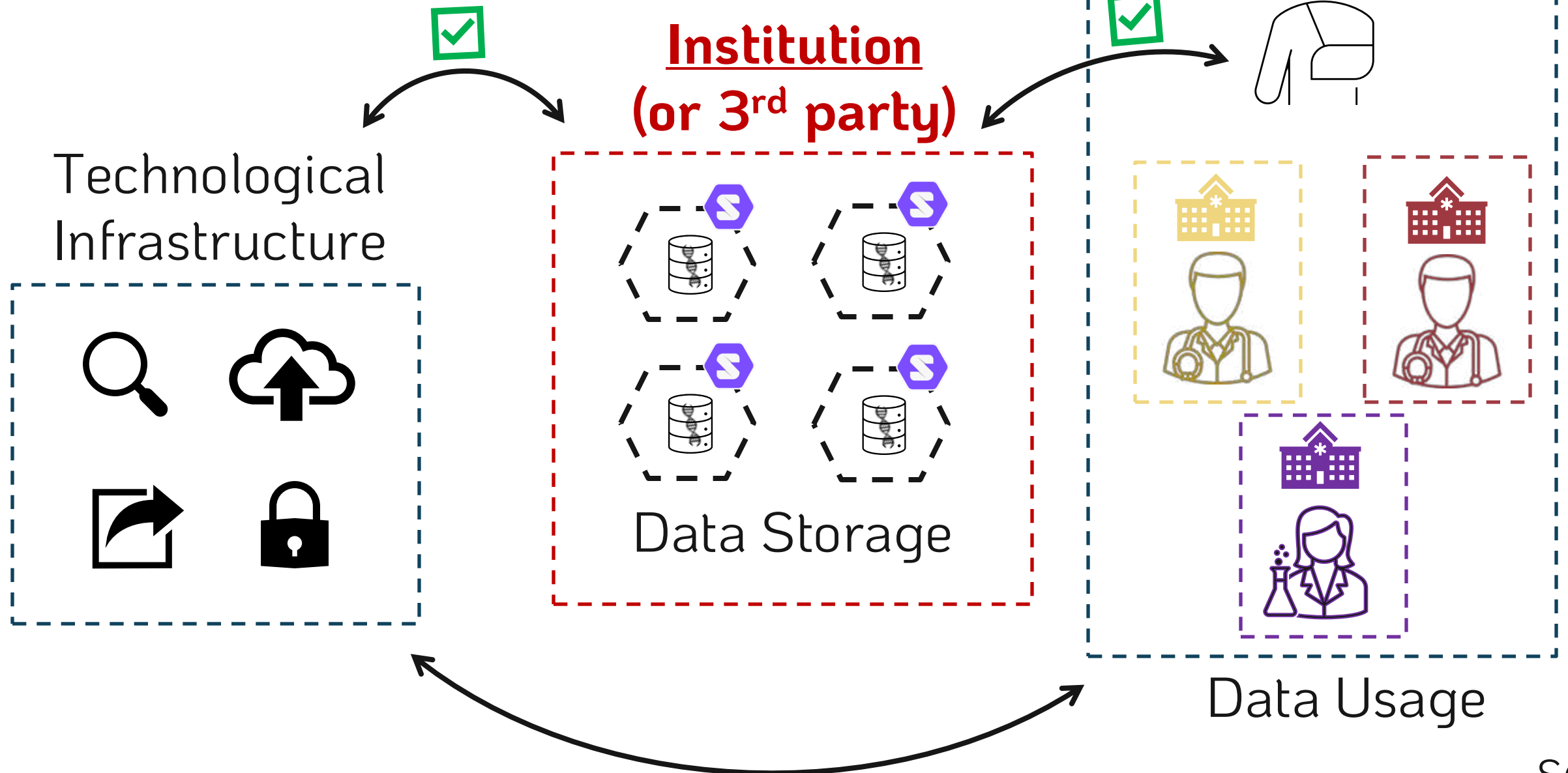
**Flanders**  
State of the Art

Flemish Smart Dataspace



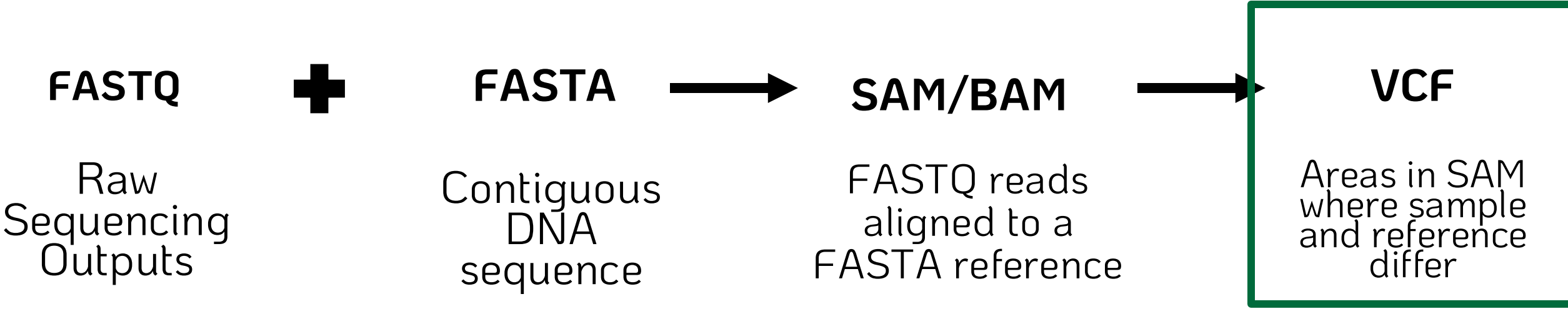
Solid Lab Use-cases

# A Production Solid Solution





# Semantic VCF – Sequencing Data Formats Intro



```
1 >gi|568336023|gb|CM000663.2| Homo sapiens chromosome 1, GRCh
2 CCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAAC
3 ACCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAA
4 TAACCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCT
5 CTAACCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCT
6 CAACCCCAACCCCAACCCCAACCCCAACCCCAACCCCTAACCCCTAACCCCTAACCC
7 CCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCC
8 TAACCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCT
9 TCTGACCTGAGGAGAACTGTGCTCCGCTTTCAGAGTACCACCGAGGAGGAGGAGG
10 CGCCCTCGCGGTGCTCTCCGGGTCTGTGCTGAGGAGAACGCAAGGAGGAGGAGGAGG
11 CGGCGCAGGCGCAGAGAGGCGCGCCGCGCGCGGCGCAGGCGCAGGCGCAGGCGCA
12 AGAGAGGCGCGCCGCGCGCGGCGCAGGCGCAGAGAGGCGCGCCGCGCGCGGCGCAGGCGCAGACACATGCTAGCGCGTGGGGGTG
13 CGGCGCGGCGCAGGCGCAGACACATGCTAGCGCGTGGGGGTG
14 GCCGCGCCGCGCAGGCGCAGAGACACATGCTACCGCGTCCAGGCGCACCGCGCGCGGCGCAGGCGCAGAGACACATGCTAGCG
15 GCAGAGACGCAAGCCTACGGGCGGGGGTTGGGGGGCGTGTG
16 GCTGGGGCGGGGGGAGGGTGGCGCGTGCACGCGCAGAACT
17 GGTAGAACCTCAGTAATCCGAAAAGCCGGGATCGACCGCC
18 GCTTGCTCACGGTGTGTGCCAGGGCGCCCTGCTGGCGAC
19 GAGTGGTGGCCAGCGCCCTGCTGGCGCGGGGCACTGCAG
```

@FORJUSP02  
CCGTCAATTC  
+  
AAAAAAAAAA

Q scores (as ASCII ch  
Base=T, Q='!' =25

```
@HD VN:1.0 SO
@SQ SN:chr20
@PG ID:TopHat
lign-edit-dist 2
20 /data/user446/n
HWI-ST1145:74:C101
CCGTGTTTAA
C BBDDCCDDCCDDDD
AS:i:-15
HWI-ST1145:74:C101
TGCTGGATCAT
G DCDDDDDDDDDD
AS:i:-16
HWI-ST1145:74:C101DACXX:7:1204:14760:4030 16 chr20 270877 50 100M * 0 0
GGCTTTATTGGTAAAAAAGGAATAGCAGATTTAATCAGAAATCCACCTGGCCAGCAGCACCAACCAGAAAGAAGGAAGAAGACAGGAAAAACCA
C DDDDDDDDDCCDDDDDDDDDEEEEEEEFFEFEGHHHFGDJJIHJJIJJJJIIIGGFJJIHIIJJJJJJJIGHHFAHGHHJHFGGHHFFDD@BB
AS:i:-11 XM:i:2 XO:i:0 XG:i:0 MD:Z:0A85G13 NM:i:2 NH:i:1
HWI-ST1145:74:C101DACXX:7:1210:11167:8699 0 chr20 271218 50 50M4700N50M * 0
0 GTGGCTCTCCACAGGAATGTTGAGGATGACATCCATGTCTGGGTGCACTTGGTCTCCGAGCAGAACATCTCAAATATGACCTCTCG
accepted_hits.sam
```

```
##fileformat=VCFv4.3
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=A,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT NA00001 NA00002 NA00003
20 14370 rs6054257 G A 29 PASS NS=3;DP=14;AF=0.5;DB;H2 GT:GQ:DP:HQ 0/0:48:1:51,51 1/0:48:8:51,51 1/1:43:5:..
20 17330 . T A 3 q10 NS=3;DP=11;AF=0.017 GT:GQ:DP:HQ 0/0:49:3:58,50 0/1:3:5:65,3 0/0:41:3
20 1110696 rs6040355 A G,T 67 PASS NS=2;DP=10;AF=0.333,0.667;AA=T;DB GT:GQ:DP:HQ 1/2:21:6:23,27 2/1:2:0:18,2 2/2:35:4
20 1230237 . T . 47 PASS NS=3;DP=13;AA=T GT:GQ:DP:HQ 0/0:54:7:56,60 0/0:48:4:51,51 0/0:61:2
20 1234567 microsat1 GTC G,GTCT 50 PASS NS=3;DP=9;AA=G GT:GQ:DP 0/1:35:4 0/2:17:2 1/1:40:3
```

# What does a VCF File Look like?

## Example

VCF header

```
##fileformat=VCFv4.0
##fileDate=20100707
##source=VCFtools
##reference=NCBI36
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality (phred score)">
##FORMAT=<ID=GL,Number=3,Type=Float,Description="Likelihoods for RR,RA,AA genotypes (R=ref,A=alt)">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##ALT=<ID=DEL,Description="Deletion">
##INFO=<ID=SVTYPE,Number=1,Type=String,Description="Type of structural variant">
##INFO=<ID=END,Number=1,Type=Integer,Description="End position of the variant">
```

Body

| #CHROM | POS | ID  | REF | ALT   | QUAL | FILTER | INFO               | FORMAT   | SAMPLE1  | SAMPLE2 |
|--------|-----|-----|-----|-------|------|--------|--------------------|----------|----------|---------|
| 1      | 1   | .   | ACG | A,AT  | .    | PASS   | .                  | GT:DP    | 1/2:13   | 0/0:29  |
| 1      | 2   | rs1 | C   | T,CT  | .    | PASS   | H2;AA=T            | GT:GQ    | 0 1:100  | 2/2:70  |
| 1      | 5   | .   | A   | G     | .    | PASS   | .                  | GT:GQ    | 1 0:77   | 1/1:95  |
| 1      | 100 | .   | T   | <DEL> | .    | PASS   | SVTYPE=DEL;END=300 | GT:GQ:DP | 1/1:12:3 | 0/0:20  |

Mandatory header lines

Optional header lines (meta-data about the annotations in the VCF body)

Reference alleles (GT=0)

Alternate alleles (GT>0 is an index to the ALT column)

Deletion

SNP

Large SV

Insertion

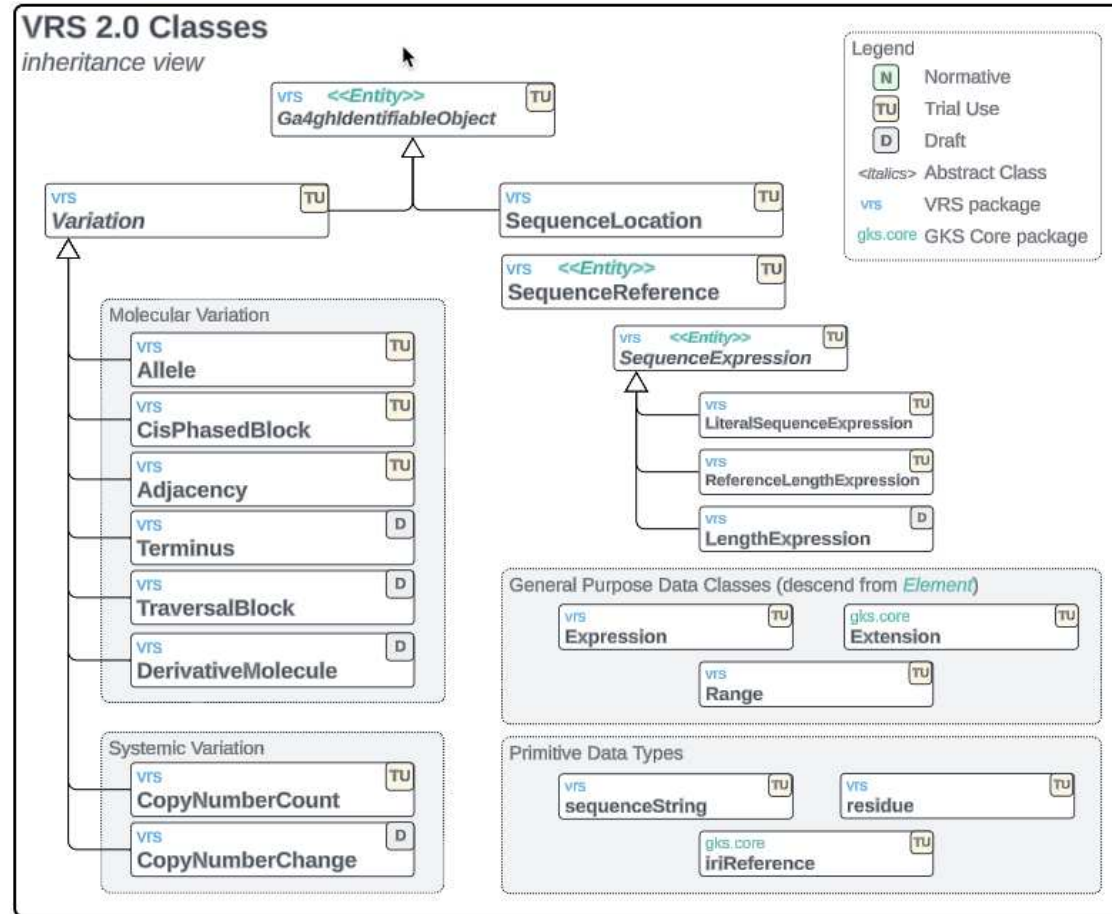
Other event

Phased data (G and C above are on the same chromosome)

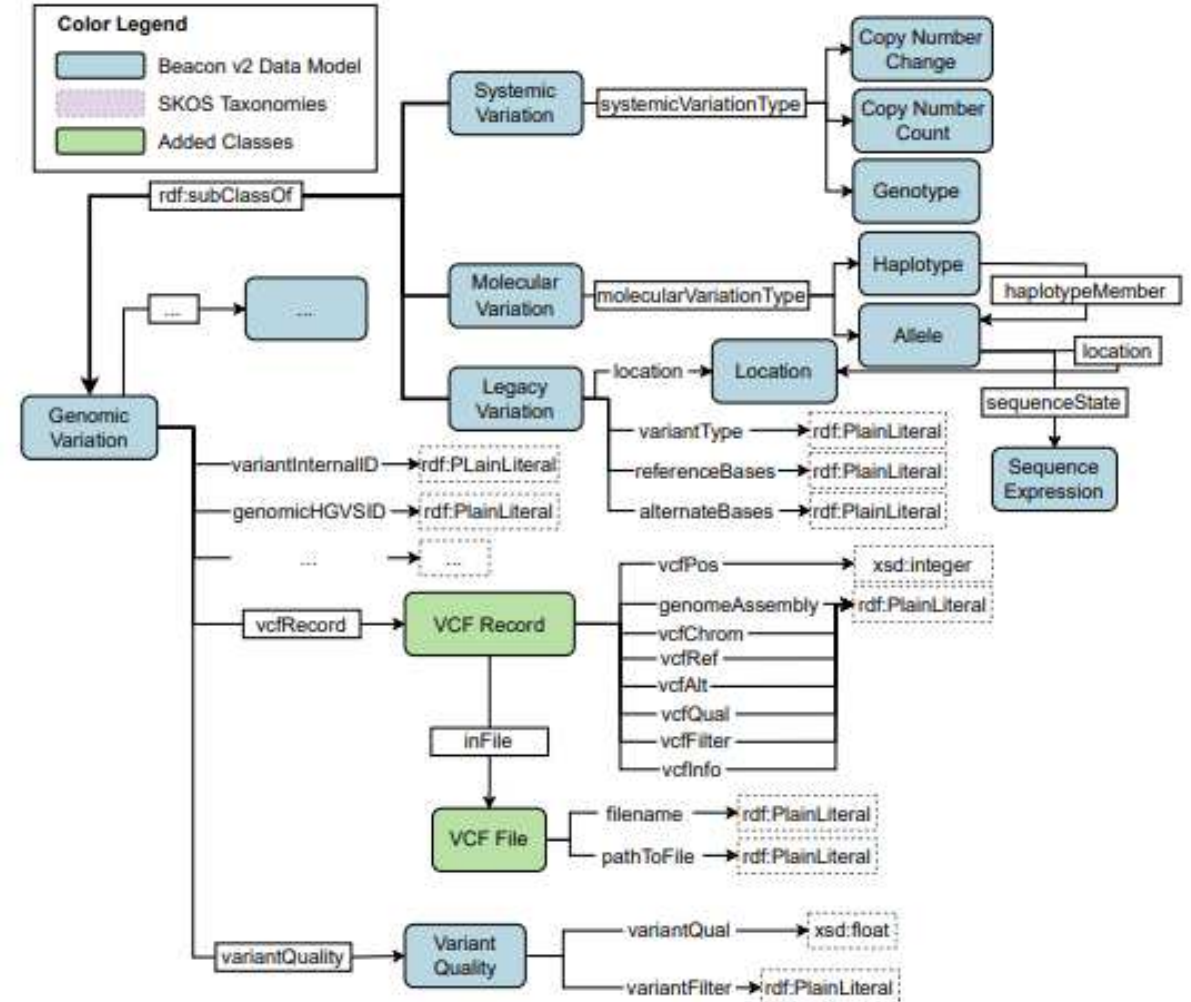


# Representing VCF Semantically?

## GA4GH Variant Schema:

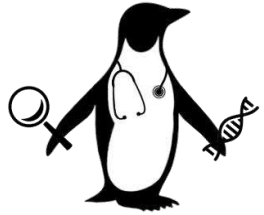


## HERO Genomics Ontology:



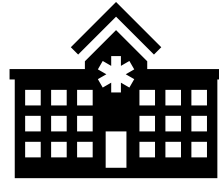
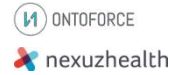


# Envisioned Product Evolution



PhD End

PhD  
Deliverable



Short Term

Start-up / Product  
development  
+  
Hospital implementation



Medical record



Medium Term

Integrate with EHR  
+  
Increase application  
domains



Long Term

Increase 3<sup>rd</sup> party  
application support  
+  
Scale to Europe and  
beyond